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# Distribution of per- and poly-fluoroalkyl substances and their precursors in human blood

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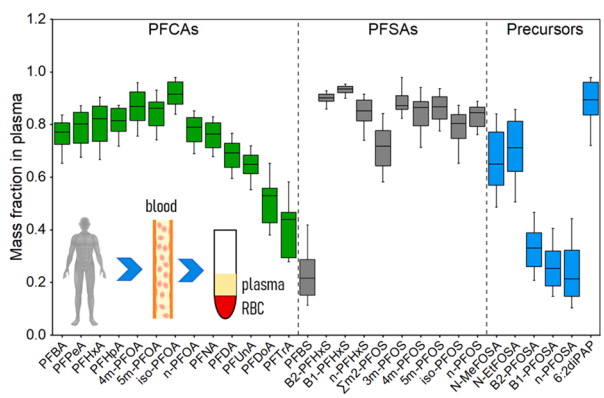
## HIGHLIGHTS

- Mean $F_p$ values of $C_4$–$C_7$ PFCAs increased with the increasing chain length.
- $C_7$–$C_{13}$ PFCAs exhibited a trend of gradually decreasing mean $F_p$ with chain length.
- This study first reports the partitioning of N-MeFOSA and N-EtFOSA in human blood.
- $F_p$ of N-MeFOSA and N-EtFOSA are comparable to PFUnA and PFDA, respectively.
- Branched isomers had higher $F_p$ than linear isomers for PFOA, PFHxS, and PFOSA.

## GRAPHICAL ABSTRACT

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## ABSTRACT

Many studies have examined per- and poly-fluoroalkyl substances (PFASs) in human blood. However, the distribution of PFASs in human blood remains not well known, especially for perfluorooctane sulfonate (PFOS) precursors. In this study, human blood samples ($n = 162$) were collected from general Chinese population, and then the isomer-specific partitioning of PFASs between human plasma and red blood cells (RBCs) were investigated. Perfluorooctanoate (PFOA) and PFOS were consistently the predominant PFASs in both human plasma and RBCs. In human blood, among $C_4$–$C_7$ perfluoroalkyl carboxylates (PFCAs), the calculated mean mass fraction in plasma ($F_p$) values increased from 0.76 to 0.82 with the increasing chain length. $C_7$–$C_{13}$ PFCAs exhibited a trend of gradually decreasing mean $F_p$ with chain length. Among PFAS precursors, 6:2 fluorotelomer phosphate diester had the highest mean $F_p$ value (0.87 ± 0.11). Calculated $F_p$ values of N-methyl perfluorooctanesulfonamide (N-MeFOSA) and N-ethyl perfluorooctanesulfonamide (N-EtFOSA) were 0.66 ± 0.13 and 0.70 ± 0.12, respectively. Individual branched isomers consistently had greater $F_p$ values than their corresponding linear isomers for PFOA, PFHxS, and perfluoroctane sulfonamide. To our knowledge, this study first

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reports the distribution of N-MeFOSA and N-EtFOSA in human blood, contributing to the better understanding of the occurrence and fate of PFASs in humans.

## 1. Introduction

Perfluoroalkyl substances (PFASs), such as perfluoroalkyl carboxylates (PFCAs) and perfluoroalkyl sulfonates (PFSAs), are a class of artificial fluorinated organic materials firstly manufactured in the early 1950s (Kwiatkowski et al., 2020; Wang et al., 2017). These chemicals have been found in a variety of consumer products, including textiles, carpets, fluoropolymers, firefighting foams, and coatings of food packaging (Feng et al., 2020; Gluge et al., 2020). Among these compounds, perfluorooctane sulfonate (PFOS) and perfluorooctanoate (PFOA) have received considerable attention from scientists and regulatory agencies, owing to their ubiquities and long persistence in environmental matrixes, humans, and wildlife (Jian et al., 2018; Podder et al., 2021; Szabo et al., 2022). Studies have revealed positive correlations between serum levels of PFOA or PFOS and adverse human health effects (Lopez-Espinosa et al., 2012; Ojo et al., 2021).

Historically, synthesis of PFASs were primary via two methods in industry: electrochemical fluorination (ECF) and telomerization (Benskin et al., 2010a). ECF process yields complex mixtures of linear and branched structural isomers. PFCA homologues with different carbon-chain length were also generated as minor impurities in ECF process. Telomerization gives rise to relatively pure straight-chain fluorinated chemicals or isopropyl branched isomers (Benskin et al., 2010b). In North America large-scale production of PFOA was through ECF before 2002, which has likely been shifted to telomerization since 2002 (Benskin et al., 2010a). However, ECF is still being used by some manufacturers in China (e.g., Wuhan, Guangzhou, and Shenyang) to produce commercial PFOA and PFOS (Jiang et al., 2015).

The most frequently detected PFASs in human serum from around the world are PFOA, perfluorohexane sulfonate (PFHxS), and PFOS (Gockener et al., 2020; Wu et al., 2017; Zhang et al., 2014). Human exposure to these perfluoroalkyl acids (PFAAs) occurs via direct intake of themselves (such as diet and inhalation of dust), as well as via exposure to other commercial fluorochemicals with subsequent biotransformation (referred to as precursors). Their measured concentrations were always much lower than PFOA and PFOS in human blood (Martin et al., 2010). Despite that, exposure to precursor chemicals may still contribute much to the human PFAA contamination (Nilsson et al., 2013). However, some of these precursors, such as N-methyl perfluorooctanesulfonamide (N-MeFOSA) and N-ethyl perfluorooctanesulfonamide (N-EtFOSA), have not been routinely monitored in human blood. Such data is vital to understand their human internal exposure levels and to evaluate the contribution of PFAA precursor degradation to PFAAs.

Polyfluoroalkyl phosphate diesters (diPAPs), newly identified commercial fluorotelomer-based PFASs, have been detected in human serum at concentrations of microgram per liter (Luo et al., 2022; Poothong et al., 2017). These materials are approved for application in food-contact paper packaging, lubricants for polymers, caulks, and sealants (Trier et al., 2011). Among diPAPs, 6:2 and 8:2 diPAPs were always found to be the most prevalent congeners in human blood and consumer products (Kaiser et al., 2021; Loi et al., 2013). Presences of diPAPs have been demonstrated in indoor environment (De Silva et al., 2012) and drinking water (Ding et al., 2012), indicating potential routes of human exposure (Chen et al., 2022). DiPAPs can be biodegraded to PFCAs in rats and are deemed as contributors to the burden of PFCAs in the human body (D'eon and Mabury, 2011). However, the partitioning of diPAPs in human blood have not been well characterized.

Distinctions in partitioning behaviors of some PFASs in human blood have been demonstrated (Jin et al., 2016; Kärrman et al., 2006; Poothong et al., 2017). For example, a great fraction (approximately 85%) of PFOS was found in human serum (or plasma), but perfluorooctane sulfonamide (PFOSA) was mainly enriched in the blood cell fraction (Kärrman et al., 2006). However, serum (or plasma) is still the most commonly selected matrix in human biomonitoring assays on PFASs, which may underestimate the relative extent of PFAS exposure. Partitioning of PFAAs in human blood has been examined in some studies (Jin et al., 2016; Poothong et al., 2017), but the partitioning behaviors of PFOS precursor chemicals, such as N-MeFOSA and N-EtFOSA, in human blood remains not well unknown.

In the present study, we collected 162 whole blood samples from a Chinese population in Quzhou, China, to investigate the presence and distribution of PFASs, especially precursor compounds of PFOS, in human blood. The influence of perfluoroalkyl chain length, branching of perfluoroalkyl chain, and polar head groups on the partitioning of PFASs between human plasma and red blood cells (RBCs) was further investigated. To our knowledge, this is the first study investigating the partitioning of N-MeFOSA and N-EtFOSA in different human blood matrixes. Results of this study contribute to the better understanding of the fate of PFASs in humans.

## 2. Materials and methods

### 2.1. Nomenclature and acronym

The PFOA and PFOS isomer nomenclatures used here have been previously used (Benskin et al., 2007). For instance, the linear, isopropyl, 5-perfluoromethyl, total dimethyl-substituted branched isomers, and sum of all isomers of PFOS are named as *n*-, *iso*-, 5*m*-, $\sum m_2$-, and PFOS, respectively. For PFHxS and PFOSA, the chromatographically separated isomers in the present study are unable to be unambiguously assigned due to the lack of authentic standards. Therefore, the various branched isomers of PFHxS and PFOSA observed in their chromatograms were assigned simply as $B_x$ (where x = 1, 2, 3, etc.) in decreasing order of retention time (e.g., $B_1$-PFOSA and $B_2$-PFOSA) (Figure S1 of the Supporting Information, SI).

### 2.2. Standards and reagents

Native and isotopically labeled PFAS standards, including brPFHxSK, brPFOSK, TPFOA, PFAC-MXB, MPFAC-MXA, PFOSA, M8FOSA-M, N-MeFOSA, N-EtFOSA, N-methyl perfluorooctanesulfonamidoacetate (N-MeFOSAA), N-ethyl perfluorooctanesulfonamidoacetate (N-EtFOSAA), 6:2 diPAP, and 8:2 diPAP were purchased from Wellington Laboratories (Guelph, ON, Canada). N-MeFOSA, N-EtFOSA, N-MeFOSAA, and N-EtFOSAA were the precursors of PFOS, while 6:2 diPAP and 8:2 diPAP were the precursors of PFCAs. ECF-derived PFOA, PFOS, and PFOSA were provided by 3 M Co. (St. Paul, MN, America). Acetonitrile, methanol, formic acid (> 99%), ammonium formate (> 98%), and HPLC-grade water were purchased from Fisher Scientific (Toronto, ON, Canada).

### 2.3. Human blood collection

Collection of whole blood samples from general population ($n = 81$) in Quzhou was conducted in October 2020. Quzhou (population ~ 2200,000; 118°01′−119°20′ E, 28°14′−29°30′ N) is located in Southeast China. All recruited participants (47 females and 34 males) are healthy general population, without evidence of occupational exposure to PFASs. The mean (range) age of female and male donors is 36 (27 −68) and 38 (30 −69) years, respectively. Detailed demographic information on the participants is provided in the SI, Table S1. Ethical clearance was

obtained from the research ethics boards of the Quzhou People's Hospital, and all participants provided informed consent prior to sampling.

Blood samples (ca. 5 mL each) were collected from participants into BD-Vacutainer® blood collection tubes (Becton Dickinson, NJ, USA) embedded with ethylene diaminetetraacetic acid as an anticoagulant. After gentle mixing, a portion of fresh whole blood (2.0 mL) was withdrawn into a 5 mL polypropene (PP) centrifuge tube (Fisher Scientific, ON, Canada), and then immediately centrifuged at 3500 g for 10 min to separate plasma and red blood cell (RBC). The RBC fraction was rinsed with pure water to remove residual plasma-associated PFASs. Hematocrit of sampled human blood was 40–59% (mean, 49%) and 42–59% (54%) for females and males, respectively. These values are within the range reported for Chinese population (Ge et al., 1997). We stored all prepared plasma and RBC samples at − 80 °C before extraction for PFASs. Field blanks consisting of Milli-Q water were also transported with collected blood samples during the sampling campaign.

### 2.4. Sample extraction

Sample Extraction. Extraction of human plasma samples were adapted from Reagen et al. (2008), but with minor modifications. Prior to extraction all samples were spiked with mass-labeled PFASs (MPFAC-MXA and $^{13}C_8$-PFOSA). Acetonitrile (5 mL) was added to prepared human plasma in a PP tube (Corning, NY, USA). The mixture was vortex mixed for 1 min, sonicated at 400 KHz for 35 min, and then shaken at 200 rpm for 20 min. The supernatant of the mixture was separated by centrifugation at 4500 g for 10 min, and then transferred to a new PP tube. Extraction was repeated once more with the same method, and the combined extract was passed through the ENVI-Carb cartridge (250 mg, 6 cc, Supelclean; Sigma-Aldrich, Canada), which was pre-conditioned with 6 mL acetonitrile. Purified extract was reduced to dryness in a gentle stream of high-purity nitrogen, and then reconstituted with 100 μL of 60% methanol/40% water for instrumental analysis.

The method of Hanssen et al. (2013) was applied with minor modifications to extract PFASs from RBC samples. In brief, after spiking with internal standards 5 mL of acetonitrile was added to the blood cells in a 10-mL PP tube. After mixing on a vortex for 2 min, the mixture was shaken at 200 rpm for 20 min. The supernatant was separated by centrifugation at 3500 g for 10 min, and then transferred to a new PP tube. The extraction procedure was repeated once more as described above. All of the organic extracts were combined and evaporated to dryness under gentle nitrogen flow. The residual was reconstituted in 100 μL of methanol (60%)/water (40%).

### 2.5. PFAS isomer separation and analysis

PFAS Isomer Separation and Analysis. Quantification of target analytes were performed using ultra performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) with a Waters UPLC platform (ACQUITY) coupled to a Waters mass spectrometer (XEVO_TQS; Milford, MA, USA). In brief, 10 μL of the extract was injected onto an Ascentis Express F5 column (2.7 μm, 90 Å, 10 cm × 2.1 mm; Sigma-Aldrich, Canada) maintained at 40 °C. Gradient elution condition started at 60% A (water adjusted to pH 6.0 with formic acid) and 40% B (HPLC-grade methanol), which was held for 1 min, ramped to 60% B by 15 min, increased to 85% B by 18 min, to 100% B by 22 min, held until 24 min, and finally returned to the initial condition by 25 min. Mass spectra data were collected using negative electrospray ionization and multiple reaction monitoring. Ions used for quantification and confirmation are shown in the SI (Table S2).

### 2.6. Mass fraction calculation

Mass Fraction Calculation. Partitioning potential of PFASs between human plasma and RBCs was evaluated by calculating their mass fraction in plasma ($F_p$) (Jin et al., 2016), using the following equation:

$$F_P = \frac{(1 - H_C) \times C_P}{H_C \times C_{RBC} + (1 - H_C) \times C_P}$$

where $C_P$ and $C_{RBC}$ refer to the concentrations of individual PFAS in human plasma (ng/mL) and RBC (ng/mL), respectively. $H_c$ means the hematocrit. It is unitless, and was directly measured using the automatic hematology analyzer (BC-5000, Mindray; Wuhan, China).

### 2.7. QA/QC

One blank sample (HPLC-grade water) was extracted with each batch of ten samples to monitor background contamination. One solvent blank (methanol) was injected after every ten samples to monitor any carry-over. Quantification of all target analytes were performed using internal standard calibration curves (not matrix-matched) with six points, which were highly linear ($R^2 > 0.995$). The limit of detection (LOD) was defined as the mean PFAS concentration of procedural blanks ($n = 3$) plus three times the standard deviation of the blank responses. If the specific PFAS was not detected in procedural blanks, the LOD was defined as the concentration with a signal-to-noise ratio of 3. The LOD of PFASs was in the range of 15 − 93 pg/mL in human plasma and 38 − 92 pg/mL in RBCs (SI, Table S3). Extraction recoveries were assessed by adding known amounts of target compounds to human plasma ($n = 5$) and RBCs ($n = 5$), and were in the rage of 76 − 103% and 76 − 100%, respectively (SI, Table S3). Extraction recovery of branched isomers of PFHxS and PFOSA were 86 − 99% and 90 − 104%, respectively. Recoveries were comparatively low for the extraction of PFTeA (79 ± 6.3%) and 8:2 diPAP (76 ± 4.9%) from plasma. This is possibly due to the lack of isotopically labeled internal standards for these compounds. The endogenous interference-free transition ($m/z$ 399/119) was chosen to confirm the PFHxS quantification (Chan et al., 2009). Isomer profiles of PFHxS, PFOA, and PFOS were all conserved during extraction in the recovery experiment, indicating the matrix did not influence isomer profiles, as previously reported (Benskin et al., 2010b).

### 2.8. Statistical analysis

Statistical Analysis. Difference with $p < 0.05$ was considered statistically significant, and all statistical analysis was conducted using IBM SPSS® Statistics (Armonk, NY, USA). For statistical analysis, concentrations of PFAS lower than LODs were replaced with LODs/$\sqrt{2}$. Only concentrations of PFASs higher than LODs were used to calculate $F_p$ values. Spearman's correlation analysis was used to assess the relationships between concentrations of individual PFASs in human serum or RBCs, since most of these data exhibited skewed distribution (Shapiro-Wilks test, $p < 0.05$). Mann-Whitney $U$ test was applied to evaluate the difference in human plasma or RBC PFAS concentrations between male and female participants, and the difference in human plasma or RBC concentrations between individual PFASs.

## 3. Results and discussion

### 3.1. PFASs in human plasma and RBCs

In human plasma ($n = 162$), all target PFASs, except PFTeA and N-EtFOSAA, were detected, with the total PFAS ($\sum$PFAS) concentration of 2.7 − 23 ng/L. PFOA (0.86–11 ng/mL), PFNA (0.15 −1.3 ng/mL) and PFDA (0.10 −1.9 ng/mL) were the predominant PFCAs in human plasma (Table 1). Among PFSAs, PFOS (0.49–14 ng/mL) displayed higher plasma levels than PFHxS (0.36–3.2 ng/mL) and PFBS (<LOD–0.13 ng/mL). This congener profile (Fig. 1) is similar to those reported for elsewhere in China (e.g., Shanghai, Tianjin, and Shijiazhuang) (Wu et al., 2017; Zhang et al., 2013b; Zhang et al., 2014).

Four PFOS precursors (i.e., PFOSA, N-MeFOSA, N-EtFOSA, and N-

**Table 1**
Isomer-Specific Concentrations (ng/mL) of PFSAs in Human Plasma and RBC Samples. NC Means Not Calculated.

| | Plasma ($n$ = 162) | | | | RBC ($n$ = 162) | | | |
|---|---|---|---|---|---|---|---|---|
| | DF | Mean | Min | Max | DF | Mean | Min | Max |
| **PFBA** | 78% | 0.37 | <LOD | 0.65 | 69% | 0.11 | <LOD | 0.31 |
| **PFPeA** | 57% | 0.22 | <LOD | 0.83 | 51% | 0.13 | <LOD | 0.74 |
| **PFHxA** | 100% | 0.42 | 0.20 | 0.88 | 100% | 0.18 | 0.086 | 0.90 |
| **PFHpA** | 43% | NC | <LOD | 0.72 | 40% | NC | <LOD | 0.62 |
| ***n*-PFOA** | 100% | 4.0 | 0.77 | 9.9 | 100% | 1.1 | 0.15 | 3.9 |
| ***iso*-PFOA** | 100% | 0.15 | 0.087 | 0.89 | 90% | 0.098 | <LOD | 0.25 |
| **5*m*-PFOA** | 72% | 0.11 | <LOD | 0.71 | 67% | 0.074 | <LOD | 0.18 |
| **4*m*-PFOA** | 48% | NC | <LOD | 0.54 | 33% | NC | <LOD | 0.14 |
| **3*m*-PFOA** | 20% | NC | <LOD | 0.56 | 0% | NC | <LOD | <LOD |
| **PFNA** | 100% | 0.39 | 0.15 | 1.3 | 100% | 0.17 | 0.13 | 0.49 |
| **PFDA** | 100% | 0.42 | 0.10 | 1.9 | 100% | 0.28 | 0.15 | 0.85 |
| **PFUnA** | 100% | 0.38 | 0.27 | 1.4 | 100% | 0.30 | 0.11 | 0.57 |
| **PFDoA** | 60% | 0.15 | <LOD | 0.61 | 65% | 0.18 | <LOD | 0.99 |
| **PFTrA** | 56% | 0.10 | <LOD | 0.35 | 72% | 0.19 | <LOD | 0.62 |
| **PFTeA** | 0% | NC | <LOD | <LOD | 47% | NC | <LOD | 0.74 |
| **PFBS** | 63% | 0.11 | <LOD | 0.31 | 81% | 0.31 | <LOD | 1.2 |
| ***n*-PFHxS** | 100% | 0.50 | 0.22 | 3.0 | 94% | 0.094 | <LOD | 0.57 |
| **$B_1$-PFHxS** | 88% | 0.16 | <LOD | 0.28 | 89% | 0.046 | <LOD | 0.30 |
| **$B_2$-PFHxS** | 79% | 0.12 | <LOD | 0.15 | 77% | 0.045 | <LOD | 0.48 |
| ***n*-PFOS** | 100% | 2.6 | 0.32 | 10 | 100% | 0.40 | 0.17 | 1.8 |
| ***iso*-PFOS** | 100% | 0.64 | 0.20 | 1.6 | 100% | 0.14 | 0.074 | 0.93 |
| **5*m*-PFOS** | 100% | 0.60 | 0.095 | 1.8 | 93% | 0.12 | <LOD | 0.77 |
| **4*m*-PFOS** | 98% | 0.53 | <LOD | 1.2 | 53% | 0.073 | <LOD | 0.46 |
| **3*m*-PFOS** | 89% | 0.47 | <LOD | 0.087 | 48% | NC | <LOD | 0.39 |
| **$\sum m_2$-PFOS** | 75% | 0.094 | <LOD | 0.41 | 39% | NC | <LOD | 0.25 |
| ***n*-PFOSA** | 92% | 0.042 | <LOD | 0.11 | 100% | 0.25 | <LOD | 1.2 |
| **$B_1$-PFOSA** | 90% | 0.027 | <LOD | 0.072 | 91% | 0.095 | <LOD | 0.37 |
| **$B_2$-PFOSA** | 86% | 0.021 | <LOD | 0.058 | 89% | 0.083 | <LOD | 0.15 |
| **N-MeFOSA** | 86% | 0.32 | <LOD | 1.2 | 88% | 0.24 | <LOD | 2.7 |
| **N-EtFOSA** | 75% | 0.19 | <LOD | 0.72 | 72% | 0.14 | <LOD | 1.6 |
| **N-MeFOSAA** | 78% | 0.17 | <LOD | 0.94 | 0% | NC | <LOD | <LOD |
| **N-EtFOSAA** | 0% | NC | <LOD | <LOD | 0% | NC | <LOD | <LOD |
| **6:2 diPAP** | 81% | 0.18 | <LOD | 0.79 | 51% | 0.058 | <LOD | 0.44 |
| **8:2 diPAP** | 70% | 0.083 | <LOD | 0.21 | 0% | NC | <LOD | <LOD |

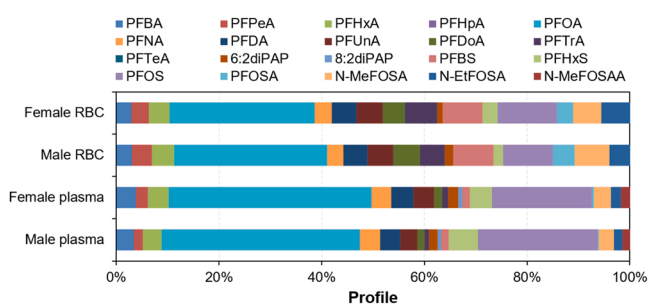


**Fig. 1.** Concentration profile of PFASs in human plasma and RBC.

MeFOSAA) were frequently detectable (75–92%) at the mean concentrations ranging from a few tens to several hundred pg/mL in human plasma. This suggests that participants may expose to these chemicals directly (as residuals in consumer products) (Zhao et al., 2022), or to PFOSA-based commercial products with subsequent metabolism (Benskin et al., 2012; Martin et al., 2010). Among them, N-MeFOSA had the highest mean concentration (0.32 ng/mL), which was 6 times higher than PFOSA (0.051 ng/mL). N-MeFOSAA was detected in 78% of human plasma, with a relatively low level (mean 0.17 ng/mL), but N-EtFOSAA was not detected at all. This may suggest that these PFOS precursors were less frequently applied in China.

The 6:2 diPAP was detected at concentrations of <LOD− 0.79 ng/mL (mean 0.18 ng/mL) in human plasma. This is similar to the plasma concentration of PFPeA (0.22 ng/mL). The 8:2 diPAP (mean 0.038 ng/mL, <LOD−0.21 ng/mL) was also detected, but less frequently (70%). The Quzhou plasma level of 6:2 diPAP exceeds that from American (<LOD−0.056 ng/mL) (Lee and Mabury, 2011), Norwegian (<0.018 −0.32 ng/mL) (Poothong et al., 2017), and Germany (<0.002 −0.067 ng/mL) (Yeung et al., 2013). Similar levels of 8:2 diPAP had been observed in serum samples from American (<LOD−0.16 ng/mL) (Lee and Mabury, 2011).

In human RBCs ($n$ = 81), all target PFASs were measurable, except N-MeFOSAA, N-EtFOSAA, and 8:2 diPAP. The predominant PFCA in human RBCs was PFOA (0.35–4.2 ng/mL), followed by PFUnA (0.11–0.57 ng/mL) and PFDA (0.15–0.85 ng/mL). PFOS (0.23–2.1 ng/mL) was still the major PFSA in human RBCs. Among detected PFOS precursors, PFOSA was predominant with a mean concentration of 0.32 ng/mL, followed by N-MeFOSA (0.24 ng/mL, <LOD–2.7 ng/mL) and N-EtFOSA (0.14 ng/mL, <LOD–1.6 ng/mL). The 6:2 diPAP was less frequently (51%) detected with lower concentrations in RBCs (<LOD–0.44 ng/mL), compared with that in plasma.

Significantly positive correlations were shown among detected PFAS precursor compounds (e.g., 6:2 diPAP, PFOSA, and N-MeFOSA) (SI, Table S4 and S5) in both human plasma and RBCs, suggesting that these chemicals may have common sources. However, PFOA and PFOS were always significantly and positively correlated with $C_7$ −$C_{13}$ PFCAs and PFHxS ($C_6$ PFSA), but scarcely with their respective precursors in human plasma and RBC samples. This suggests that PFCAs and PFSAs in human plasma were predominantly originated from the direct exposure, but less from biodegradation of precursor compounds. Concentrations of most individual PFASs in human plasma were positively correlated with those in RBCs ($r_s$ = 0.43–0.90, $p$ < 0.05).

### 3.2. Age- and gender-related differences

Concentrations of PFHxS and PFOS in male plasma were significantly ($p$ = 0.048 and 0.026, respectively) higher than that in females. A similar trend was found for PFOSA in human plasma, which was

significantly higher ($p = 0.022$) in males (mean 0.051 ng/mL) than in females (0.034 ng/mL). No statistical difference between genders was found for other PFASs. In RBCs, the significant gender-specific difference in concentration was only found for PFHxS, which was higher ($p = 0.047$) in males than in females, consistent with the finding in plasma. These results are consistent with previous results (Ingelido et al., 2020; Zhang et al., 2014).

Concentrations of $C_7$ –$C_{13}$ PFCAs ($r_s = 0.31$ –0.63, $p < 0.01$), PFHxS ($r_s = 0.26$, $p < 0.017$), PFOS ($r_s = 0.50$, $p < 0.01$), and N-MeFOSAA ($r_s = 0.29$, $p < 0.023$) in plasma were significantly increased with age, in line with previous results (Duffek et al., 2020; Zhang et al., 2013b). In RBCs, an increasing trend with age was also observed for $C_8$ –$C_{11}$ PFCAs ($r_s = 0.30$ –0.54, $p < 0.01$), PFHxS ($r_s = 0.25$, $p = 0.024$), and PFOS ($r_s = 0.45$, $p < 0.01$).

### 3.3. PFAS isomer profiles in human plasma and RBCs

In human plasma and RBC samples, PFOA, PFHxS, PFOS, and PFOSA were detected with various branched isomers (Table 1 and Fig. 2). Among branched PFOA isomers, *iso*-PFOA was always the predominant isomer, and 3*m*-PFOA was less frequently detected in human plasma and RBCs. The percentage of *n*-PFOA to total PFOA (%*n*-PFOA) was significantly (paired *t*-test, $p < 0.01$) higher in RBCs (mean ± SD, 98 ± 1.4%) than that in paired plasma (95 ± 1.8%), and both of which greatly exceeded that in ECF PFOA (mean 79%, range 78.6–79.2%) from 3 M Company and in commercial PFOA products (77–78%) in China (Jiang et al., 2015). This is in line with previous biomonitoring studies (Jin et al., 2020; Zhang et al., 2013b). One possibility is that humans are exposed to abundant linear PFOA precursors, such as 8:2 diPAP, as observed in this study. No statistical ($p = 0.16$) distinction in %*n*-PFOA was observed between males (95.3 ± 1.9%) and females (94.7 ± 1.8%).

Four branched PFOS isomers were detected, and *iso*-PFOS was always the most abundant. There was no statistical ($p = 0.28$) difference between the percentage of *n*-PFOS to total PFOS (%*n*-PFOS) in human plasma and RBCs. Females (mean 76%) displayed significantly higher ($p < 0.01$) %*n*-PFOS than males (71%) in plasma samples. This may be because females preferentially excreted branched PFOS isomers through some special routes, such as menstruation, pregnancy, and lactation (Beesoon et al., 2011). In addition, the mean %*n*-PFOS in Quzhou plasma (74%) was higher than that in the ECF PFOS from 3 M Company (mean 70%, range 69–71%) and in ECF PFOS products (66–72%) manufactured in China (Beesoon et al., 2011). This is possibly because *n*-PFOS is preferentially retained in humans (Zhang et al., 2013a). Because the isomer profile of PFOS is similar to that in ECF PFOS, it is unlikely that there is a significant contribution of precursor biotransformation to PFOS in human blood in this study.

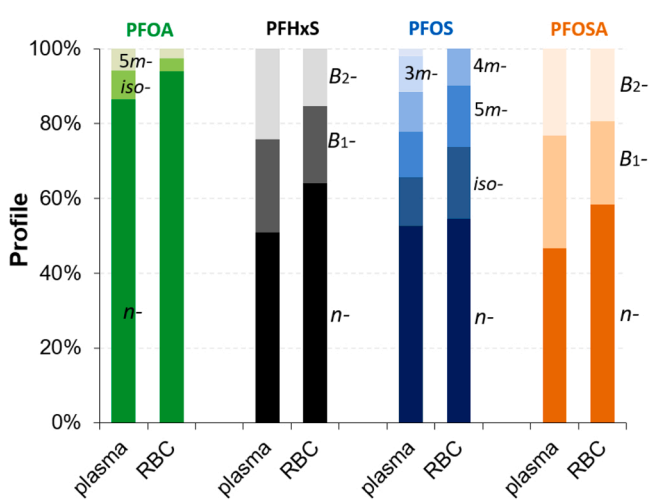


**Fig. 2.** Isomer-specific profile of PFOA, PFHxS, PFOS, and PFOSA in human plasma and RBC.

Linear PFHxS contributed 72 ± 6.9% (mean ± SD) of PFHxS in human plasma and 87 ± 4.7% in RBC samples. The %*n*-PFHxS was consistently higher in male plasma ($p = 0.002$) and RBCs ($p = 0.041$) than that in females. Gyllenhammar et al. (2018) had reported that mean %branched PFHxS in infants and their mothers from Sweden were 6.9% and 5.7%, respectively. Median %branched PFHxS in women and their infants from Norway was 12–30% (Varsi et al., 2022).

In human plasma, PFOSA contained significant quantities of branched isomers (SI, Figure S1), and *n*-PFOSA constituted 55 ± 7.2% of total PFOSA. While, in RBCs, *n*-PFOSA accounted for mean 60 ± 8.3% of total PFOSA. There was no statistical distinction ($p = 0.20$–0.66) in %*n*-PFOSA between males and females in all samples. PFOSA used in many consumer products usually contains 30% of branched isomers (Boulanger et al., 2005). Zhang et al. (2013a) had reported that opposite to PFOS human excreted *n*-PFOSA more rapidly than branched PFOSA via urine. This may lead to the deficiency of *n*-PFOSA in human plasma.

### 3.4. PFAS distribution in human blood

Calculated $F_p$ values of linear $C_4$–$C_{13}$ PFCAs ranged from 0.41 ± 0.064 (PFTrA) to 0.82 ± 0.13 (PFHpA) (Fig. 3 and SI, Table S6). No obvious gender- and age-specific distinctions in $F_p$ were observed for measured PFASs. Among $C_4$–$C_7$ PFCAs, mean $F_p$ values increased from 0.76 to 0.82 with the increasing perfluoroalkyl chain length. This is likely due to the stronger binding affinity with human serum albumin (HSA), as a result of enhanced hydrophobicity (Hebert and MacManus-Spencer, 2010). While $C_7$–$C_{13}$ PFCAs exhibited a trend of decreasing mean $F_p$ with the chain length. PFCAs with perfluorinated carbons more than eight become increasingly rigid and begin to adopt a helical geometry, which reduces their binding affinity with HSA (D'eon et al., 2010; Ellis et al., 2004). $F_p$ values of $C_8$–$C_{11}$ PFCAs (0.65–0.78) are slightly lower than that reported by Jin et al. (2016) (0.69–0.86). Poothong et al. (2017) reported that the median plasma to whole blood concentration ratios of $C_8$–$C_{11}$ PFCAs were 1.0–1.9, with no obvious changing trends observed for PFCAs.

The *n*-PFHxS (mean ± SD, 0.84 ± 0.077) had a higher mean $F_p$ than *n*-PFOS (0.83 ± 0.075) and *n*-PFOA (0.78 ± 0.073), consistent with the previous finding (Jin et al., 2016). Comparatively, PFBS had a markedly low $F_p$, with the mean of 0.24 (range 0.14–0.52), indicating a preferential partitioning to the RBC fraction. This may be because the binding strength of PFBS to HSA was an order of magnitude lower than that of PFOA, PFHxS, and PFOS (Chen and Guo, 2009; Jones et al., 2003).

Among monitored PFASs, 6:2 diPAP had the highest mean $F_p$ value (mean ± SD, 0.87 ± 0.11), indicating its preferential partitioning in the

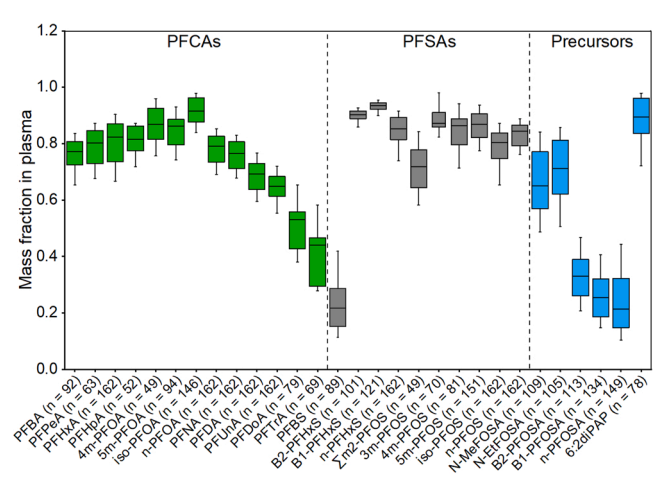


**Fig. 3.** Calculated isomer-specific $F_p$ values of PFASs in human blood.

plasma fraction. Similarly, Poothong et al. (2017) reported that the median plasma to whole blood concentration ratio of 6:2 diPAP was 2.0. We speculate that 6:2 diPAP may have a very strong affinity with HSA, owing to its two linear polyfluoroalkyl tails (i.e., $C_6F_{13}CH_2CH_2$-), since each tail could strongly bind to HSA.

*n*-PFOSA was found predominantly accumulated in blood cell fraction, with a calculated mean $F_p$ of 0.24 (0.11–0.47). This may be due to the high binding affinity of PFOSA with RBCs, as proposed by previous *in-vivo* and in-vitro results (Jin et al., 2016; Ross et al., 2012). For PFOSA, the electron of the sulfonamidate anion ($SO_2NH^-$) is stabilized by resonance (Hanssen et al., 2013), which results in the very weak binding strength to HSA. Therefore, it is not surprising that methyl and ethyl derivatives of PFOSA had much higher $F_p$ values than PFOSA. The calculated $F_p$ of N-MeFOSA and N-EtFOSA was 0.66 ± 0.13 and 0.70 ± 0.12, respectively. N-MeFOSA (SPARC; $pK_a = 6.5$) and N-EtFOSA (6.8) have much higher acid dissociation constants ($pK_a$) than PFOS (0.14) (Ahrens et al., 2012). At physiological pH, less amounts of N-MeFOSA and N-EtFOSA are negatively charged than PFOS, thereby reducing their binding with HSA, due to lower electrostatic interactions.

At the isomer-specific level, individual branched isomers consistently had greater $F_p$ values than their corresponding linear isomers for PFOA, PFHxS, and PFOSA. For instance, mean $F_p$ values of individual PFOA isomers followed the rank-order of *iso*- (0.91) > 4*m*- (0.87) > 5*m*- (0.84) > *n*-PFOA (0.78). For PFOS isomers, a slightly different pattern emerged, whereby 3*m*- (0.88), 4*m*- (0.85), and 5*m*-PFOS (0.86) had higher mean $F_p$ values than *n*-PFOS, but *iso*- (0.78; $p = 0.10$) and $\sum m_2$-PFOS (0.71; $p < 0.01$) had lower mean $F_p$ values than *n*-PFOS (0.83). In general, these results agree with their partitioning behaviors reported in rat blood (Jin et al., 2016). Branched isomers are considered more hydrophilic than the linear isomer, and are expected to have weaker affinity with HSA. One possibility of this discrepancy is that the branching of fluorocarbon tail could form a U-shape conformation to better fit the binding cavity in HSA (Luebker et al., 2002).

Interactions of PFASs with RBCs may also influence the partitioning of PFASs in human blood. Based on drug-RBC partitioning mechanisms, PFCAs and PFSAs (almost completely ionized in human blood) could bind to the phospholipid layer of RBCs, which was mainly driven by hydrophobicity. This can partially explain the decreasing Fp with hydrophobicity observed for C7–$C_{13}$ PFCAs. Nevertheless, Fp values of C4–$C_7$ PFCAs increased with their hydrophobicity. This suggests that their partitioning in human blood was primarily driven by their interaction with HSA. Linear PFAS isomers are more hydrophobic than their corresponding branched isomers, so linear isomers were expected more strongly bound to RBCs. This is possibly responsible to the higher Fp values of linear isomers for PFOA, PFHxS, PFOS, and PFOSA. Until now, it is still unclear whether the adsorption of PFASs to RBCs and to HSA are competitive. Moreover, neutral form PFOSA, N-MeFOSA, and N-EtFOSA (generally < 10% of their total amounts) are expected to enter RBCs by dissolving into the lipid bilayer membrane. Despite we are aware of that PFOSA has a strong interaction with RBCs, but empirical data on the binding affinity of other PFASs with RBCs are still very limited, which warrants further investigation.

## 4. Conclusion

This study showed that mean $F_p$ values of $C_4$–$C_7$ PFCAs increased with the increasing chain length. $C_7$–$C_{13}$ PFCAs exhibited a trend of gradually decreasing mean $F_p$ with chain length. At the isomer-specific level, individual branched isomers consistently had greater $F_p$ values than their corresponding linear isomers for PFOA, PFHxS, and PFOSA. Among PFAS precursors, 6:2 diPAP had the highest mean $F_p$ value (0.87 ± 0.11). Partitioning of N-MeFOSA and N-EtFOSA (two PFOS precursors) in human blood was investigated for the first time. Results of this study contribute to the better understanding of the occurrence and fate of PFASs in humans. Despite that, empirical data on the binding affinity of PFASs with RBCs are still very limited, which warrants further investigation.

## Environmental implication

This is the first study reporting the partitioning of N-MeFOSA and N-EtFOSA in different human blood matrixes. Results of this study contribute to the better understanding of the source and fate of PFASs in humans.

## CRediT authorship contribution statement

**Daxi Liu:** Resources, Conceptualization, Writing - original draft. **Bo Tang:** Formal analysis, Software. **Saisai Nie:** Formal analysis, Software. **Nan Zhao:** Investigation, Formal analysis. **Li He:** Resources, Supervision. **Jiansheng Cui:** Formal analysis, Conceptualization, Supervision, Writing - review & editing. **Weili Mao:** Resources. **Hangbiao Jin:** Formal analysis, Conceptualization, Writing - review & editing, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2022.129908.

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